

PHYS 230 - Electricity and Magnetism

Lecture 33

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

8 Sources of Magnetic Field

- Ampere's Law

9 Electromagnetic Induction

- Faraday's Law
- Lenz's Law

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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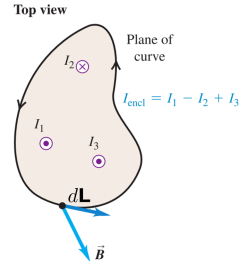
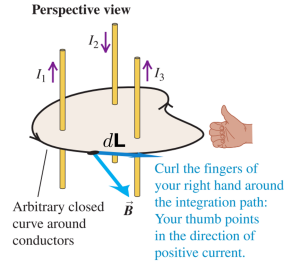
Ampere's Law

The total magnetic field $\mathbf{B}$ at any point on a closed path C that encloses a net (algebraic sum of) current $I_{\text{net-in}}$ obeys the Ampere's law

$$\oint_C \mathbf{B} \cdot d\mathbf{L} = \mu_0 I_{\text{encl}}$$

where

Figure 28.18 Ampere's law.



Ampere's law: If we calculate the line integral of the magnetic field around a closed curve, the result equals μ_0 times the total enclosed current: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{encl}}$.

Ampere's Law

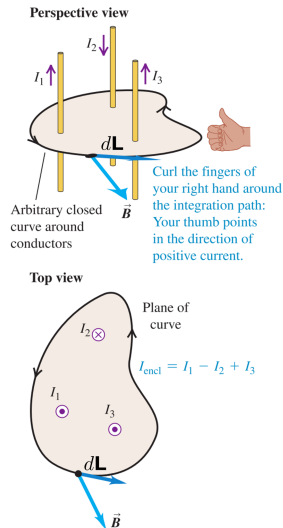
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- $d\mathbf{L}$: the infinitesimal element of path whose direction is tangent to the direction of the path, which itself is arbitrary (you choose the path direction).

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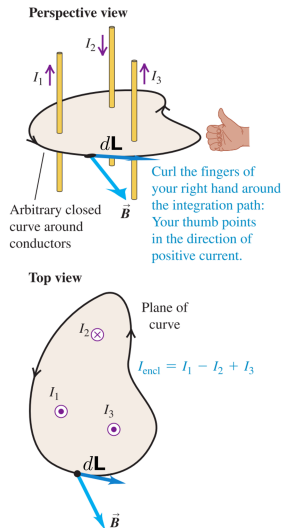
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- $d\mathbf{L}$: the infinitesimal element of path whose direction is tangent to the direction of the path, which itself is arbitrary (you choose the path direction).
- **Direction of I 's for this law:** if you align your thumb along the direction of a current I , then I is considered positive if its generated magnetic field (direction of rotation of your four fingers in the right hand rule) is in the direction of rotation of the curve of integration C . Otherwise if its generated magnetic field is in the opposite direction of rotation of the curve of integration C , then I is considered negative.

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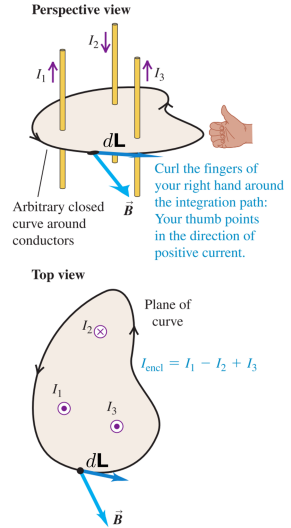
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$$\oint_C \mathbf{B} \cdot d\mathbf{L} = \mu_0 I_{\text{encl}}$$

$\oint_C \mathbf{B} \cdot d\mathbf{L} = 0$ does not necessarily mean $\mathbf{B} = 0$;
It might be that $\mathbf{B} \cdot d\mathbf{L} = 0 \Rightarrow \mathbf{B} \perp d\mathbf{L}$ along the curve C .

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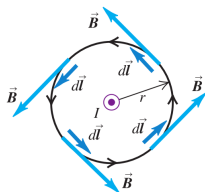
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The magnetic field a distance r from a long wire carrying a current I as in part (a) of the figure: Using Ampere's law

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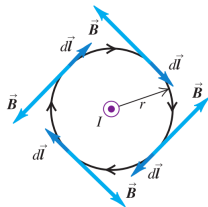
(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



(b) Same integration path as in (a), but integration goes around the circle clockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$



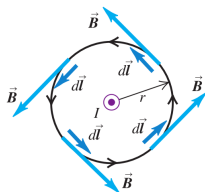
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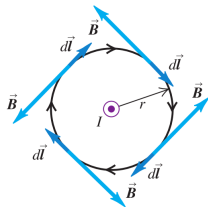
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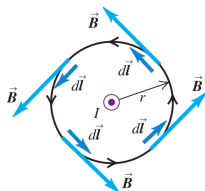
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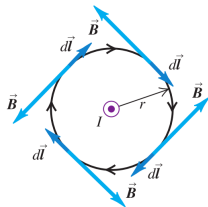
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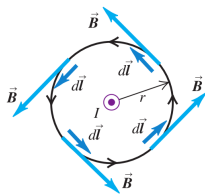
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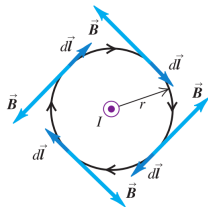
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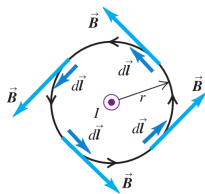
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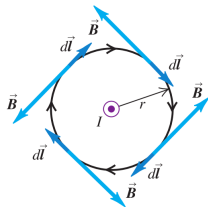
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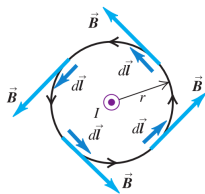
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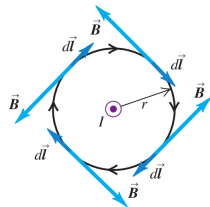
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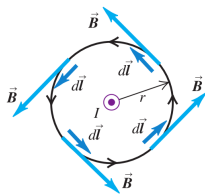
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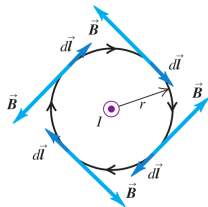
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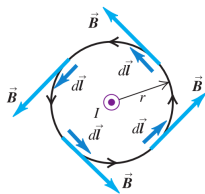
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Since the wire is axisymmetric (cylindrically symmetric), the magnitude of the magnetic field at all points on a distance r from the wire is the same so

$$B \oint_C dL = \mu_0 I$$

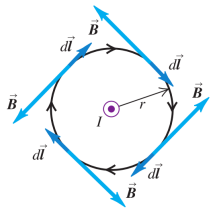
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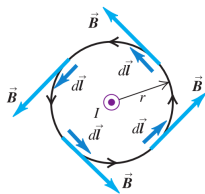
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Since the wire is axisymmetric (cylindrically symmetric), the magnitude of the magnetic field at all points on a distance r from the wire is the same so

$$B \oint_C dL = \mu_0 I \Rightarrow B(2\pi r) = \mu_0 I$$

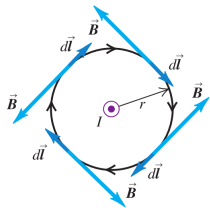
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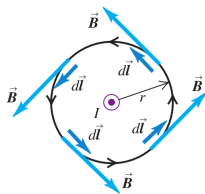
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$$B \oint_C dL = \mu_0 I \Rightarrow B(2\pi r) = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

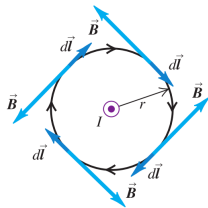
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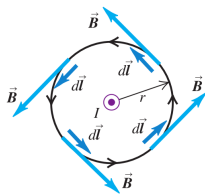
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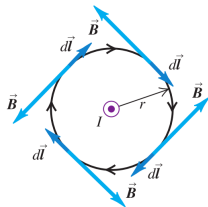
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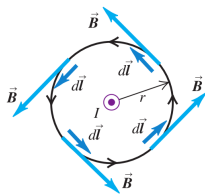
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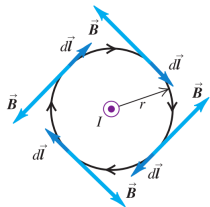
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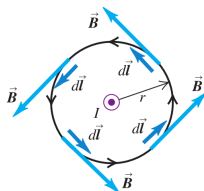
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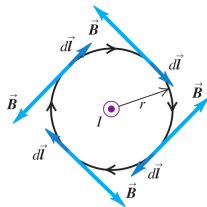
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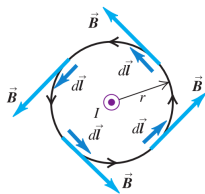
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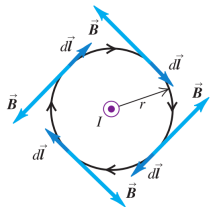
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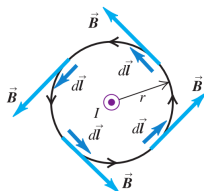
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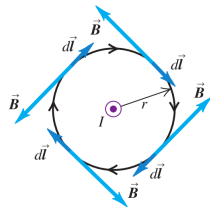
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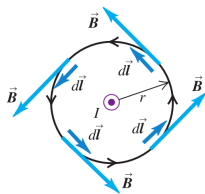
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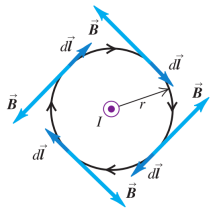
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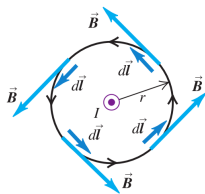
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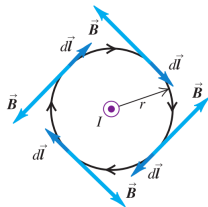
(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



(b) Same integration path as in (a), but integration goes around the circle clockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$



Ampere's Law: Example

The magnetic field a distance r from a long wire carrying a current I as in part (b) of the figure: Using Ampere's law

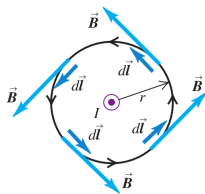
$$\oint_C \mathbf{B} \cdot d\mathbf{L} = \mu_0 I_{\text{encl}}$$
$$\oint_C (B\hat{\phi}) \cdot (dL(-\hat{\phi})) = \mu_0 (-I)$$
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Since the wire is axisymmetric (cylindrically symmetric), the magnitude of the magnetic field at all points on a distance r from the wire is the same so

$$B \oint_C dL = \mu_0 I$$

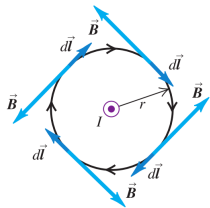
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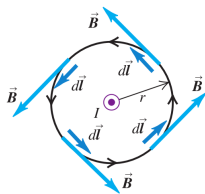
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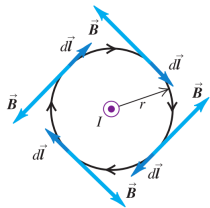
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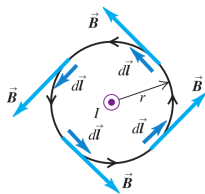
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$$B \oint_C dL = \mu_0 I \Rightarrow B(2\pi r) = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

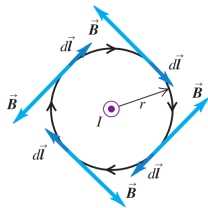
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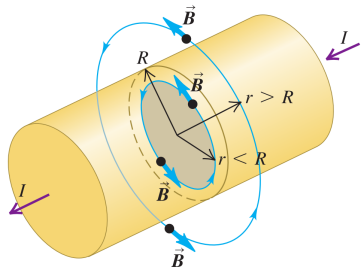
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Exercise I

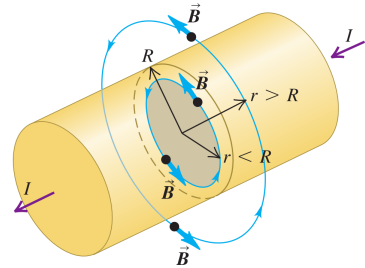
<https://epoll.srv.ualberta.ca> – Code: MBD

A cylindrical conductor with radius R carries a current I . The current is uniformly distributed over the cross-sectional area of the conductor. Find the magnetic field as a function of the distance r from the conductor axis for points inside ($r < R$) and outside ($r > R$) the conductor.



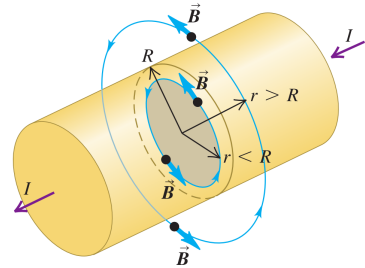
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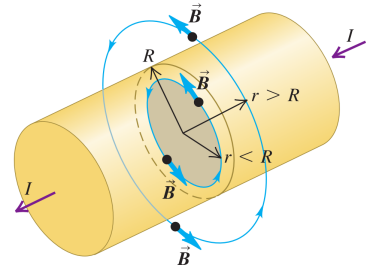
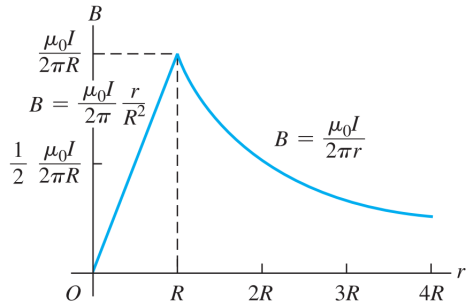
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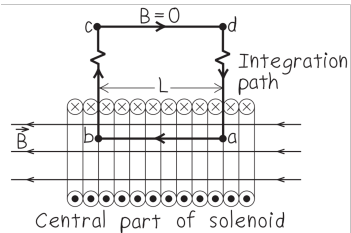
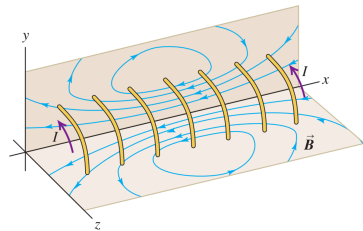
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Exercise 2

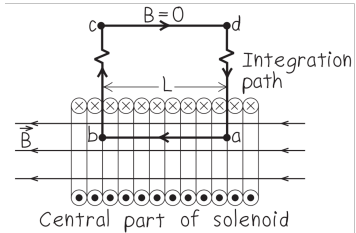
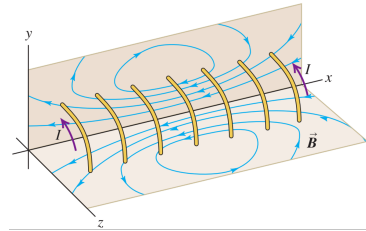
<https://epoll.srv.ualberta.ca> – Code: MBD

Use Ampere's law to find the field at or near the center of a solenoid if it has n turns per unit length and carries a current I .



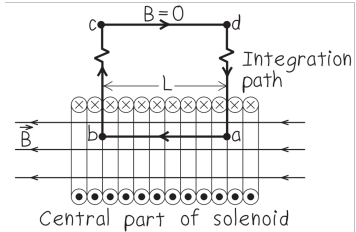
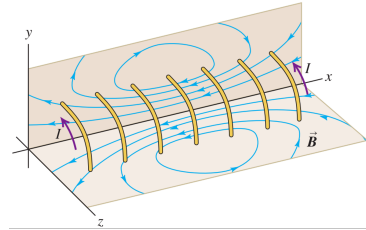
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Outline

8 Sources of Magnetic Field

- Ampere's Law

9 Electromagnetic Induction

- Faraday's Law
- Lenz's Law

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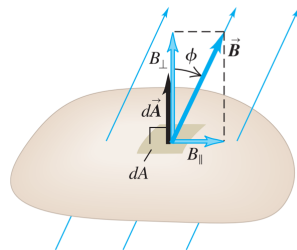
Magnetic Field Flux

As usual we define the infinitesimal flux $d\Phi_B$ of the magnetic field $\mathbf{B}$ through an infinitesimal surface $d\mathbf{A}$ as

$$d\Phi_B = \mathbf{B} \cdot d\mathbf{A}$$

where

Figure 29.3 Calculating the magnetic flux through an area element.



Magnetic flux through element of area $d\vec{A}$:

$$d\Phi_B = \vec{B} \cdot d\vec{A} = B_{\perp} dA = B dA \cos \phi$$

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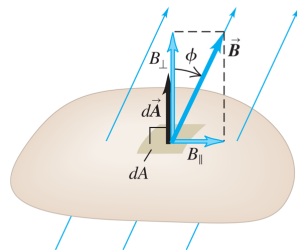
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Figure 29.3 Calculating the magnetic flux through an area element.



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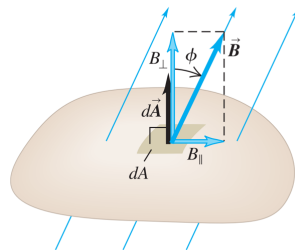
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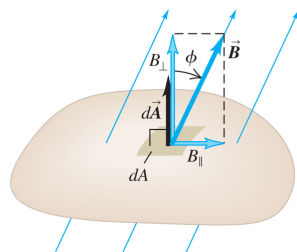
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The total flux through the finite area is

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

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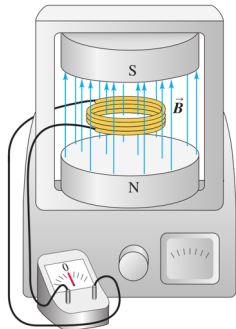
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Induced Current

Long time ago, experiments revealed that if one changes the flux

$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$ through a **closed** conducting **loop**, there will be a **current induced in the closed loop!**

Figure 29.2 A coil in a magnetic field. When the $\vec{B}$ field is constant and the shape, location, and orientation of the coil do not change, no current is induced in the coil. A current is induced when any of these factors change.



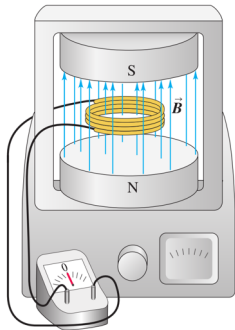
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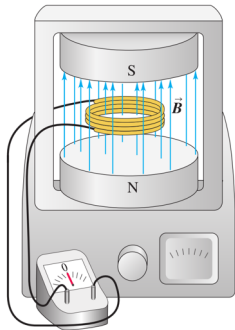
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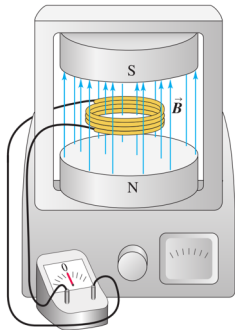
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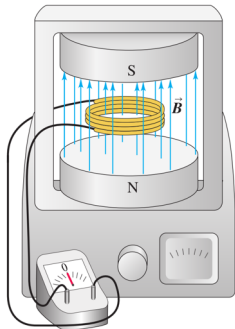
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- Change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other

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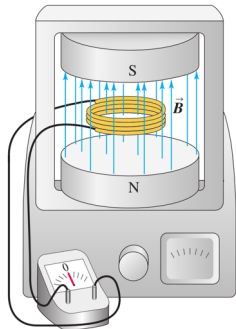


Faraday's Law

Faraday understood that the magnitude of the emf induced in a conducting loop due to change of Φ_B through it is

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

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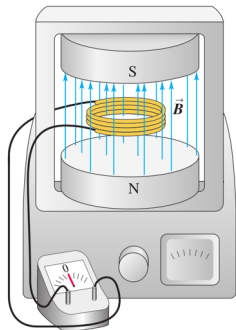
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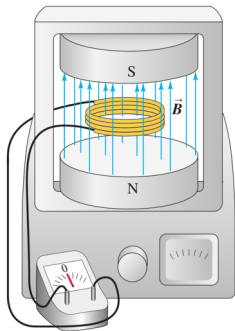
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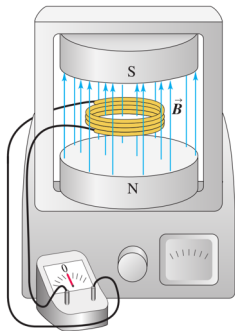
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This is the subject of Lenz's law...

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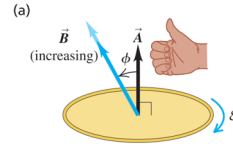
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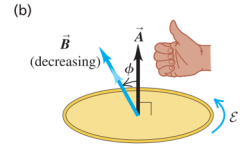
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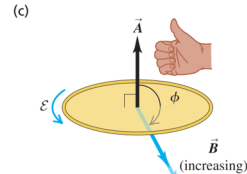
The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.



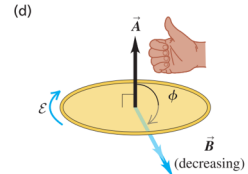
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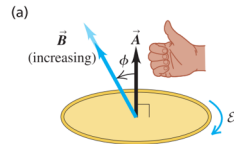


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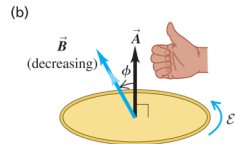
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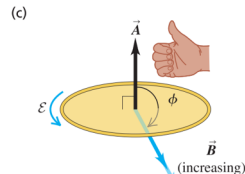
- When $\frac{d\Phi_B}{dt} \neq 0$, a current I is induced in the loop due to induced $\mathcal{E}$



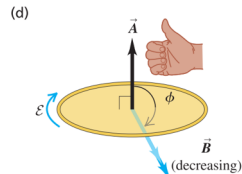
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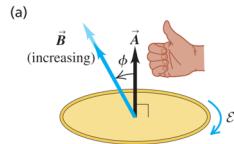


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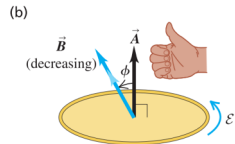
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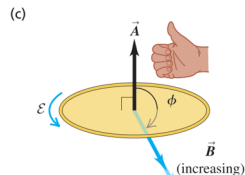
- When $\frac{d\Phi_B}{dt} \neq 0$, a current I is induced in the loop due to induced $\mathcal{E}$
- This current I , in turn, creates a new magnetic field $\mathbf{B}'$



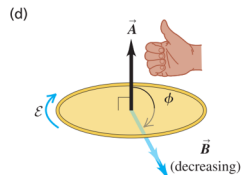
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- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

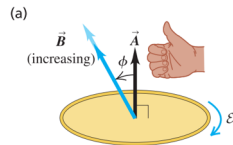


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

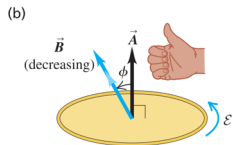
Lenz's Law

The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.

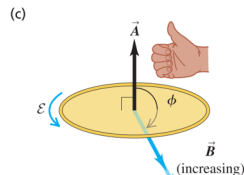
- When $\frac{d\Phi_B}{dt} \neq 0$, a current I is induced in the loop due to induced $\mathcal{E}$
- This current I , in turn, creates a new magnetic field $\mathbf{B}'$
- Direction of $\mathcal{E}$ is in the direction of induced I , such that the $\mathbf{B}'$ prevent further change of Φ_B :



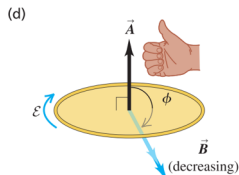
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

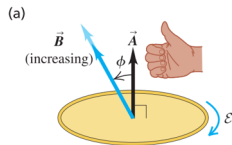


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

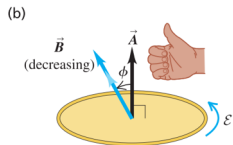
Lenz's Law

The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.

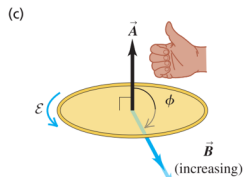
- When $\frac{d\Phi_B}{dt} \neq 0$, a current I is induced in the loop due to induced $\mathcal{E}$
- This current I , in turn, creates a new magnetic field $\mathbf{B}'$
- Direction of $\mathcal{E}$ is in the direction of induced I , such that the $\mathbf{B}'$ prevent further change of Φ_B :
- If the flux magnitude $|\Phi_B|$ is increasing, i.e., either $\frac{d\Phi_B}{dt} > 0$ and $\Phi_B > 0$ or $\frac{d\Phi_B}{dt} < 0$ and $\Phi_B < 0$, then I is such that $\mathbf{B}'$ goes against $\mathbf{B}$ to stop $|\Phi_B|$ from increasing



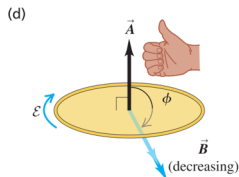
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

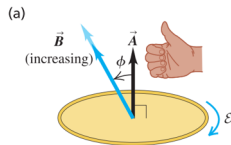


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

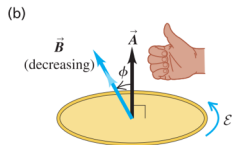
Lenz's Law

The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop.

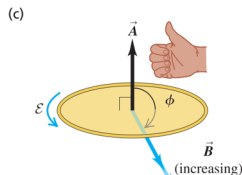
- When $\frac{d\Phi_B}{dt} \neq 0$, a current I is induced in the loop due to induced $\mathcal{E}$
- This current I , in turn, creates a new magnetic field $\mathbf{B}'$
- Direction of $\mathcal{E}$ is in the direction of induced I , such that the $\mathbf{B}'$ prevent further change of Φ_B :
- If the flux magnitude $|\Phi_B|$ is increasing, i.e., either $\frac{d\Phi_B}{dt} > 0$ and $\Phi_B > 0$ or $\frac{d\Phi_B}{dt} < 0$ and $\Phi_B < 0$, then I is such that $\mathbf{B}'$ goes against $\mathbf{B}$ to stop $|\Phi_B|$ from increasing
- If the flux magnitude $|\Phi_B|$ is decreasing, i.e., $\frac{d\Phi_B}{dt} < 0$ and $\Phi_B > 0$ or $\frac{d\Phi_B}{dt} > 0$ and $\Phi_B < 0$, then I is such that $\mathbf{B}'$ adds to $\mathbf{B}$ to stop $|\Phi_B|$ from decreasing



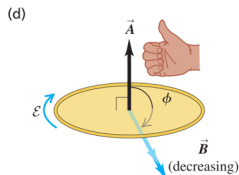
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



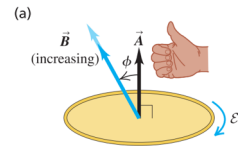
- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



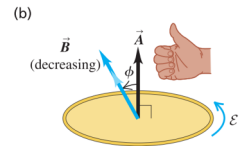
- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

Lenz's Law

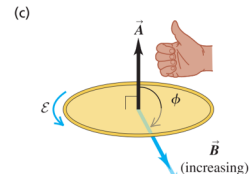
How to find the direction and sign of induced $\mathcal{E}$ and I :



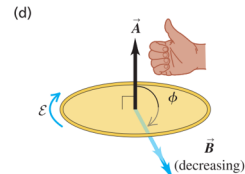
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

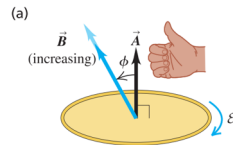


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

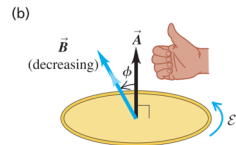
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

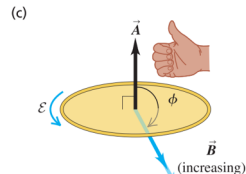
- I. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:



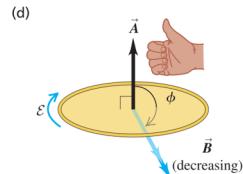
- Flux is positive ($\Phi_B > 0$) ...
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- Induced emf is negative ($\mathcal{E} < 0$).



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- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



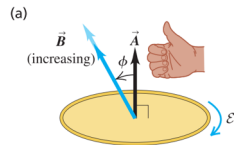
- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

Lenz's Law

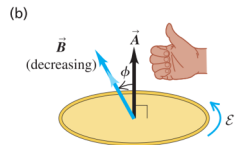
How to find the direction and sign of induced $\mathcal{E}$ and I :

I. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:

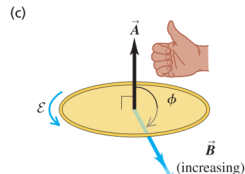
I.I $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)



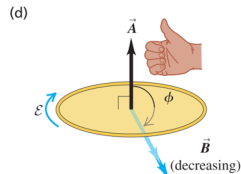
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).



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- Induced emf is positive ($\mathcal{E} > 0$).



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- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

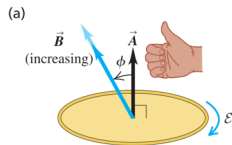
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

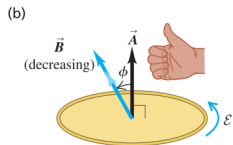
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:

1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)

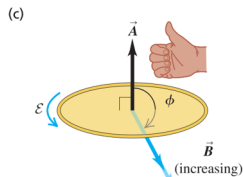
1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B



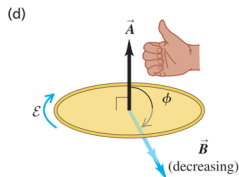
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).



- Flux is positive ($\Phi_B > 0$) ...
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- ... and becoming more negative ($d\Phi_B/dt < 0$).
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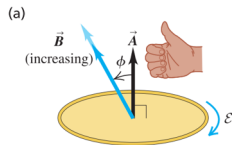


- Flux is negative ($\Phi_B < 0$) ...
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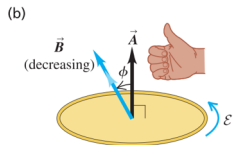
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

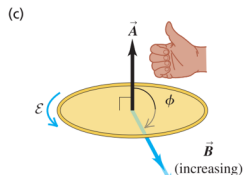
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing



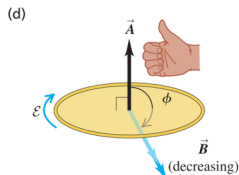
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
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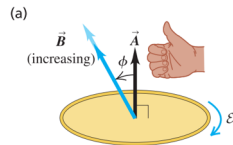


- Flux is negative ($\Phi_B < 0$) ...
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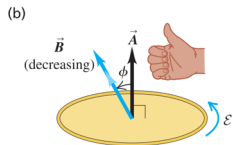
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

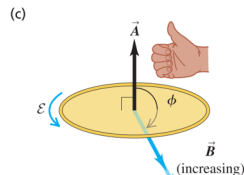
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing
3. The direction of $\mathcal{E}$ is the same as I



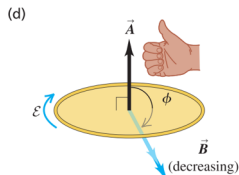
- Flux is positive ($\Phi_B > 0$) ...
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- Induced emf is negative ($\mathcal{E} < 0$).



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- ... and becoming less positive ($d\Phi_B/dt < 0$).
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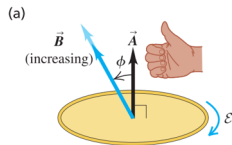


- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
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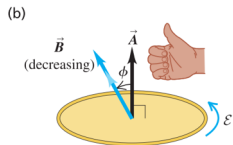
Lenz's Law

How to find the direction and sign of induced $\mathcal{E}$ and I :

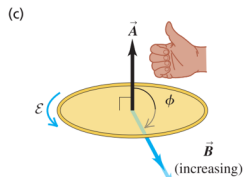
1. Check if magnitude of Φ_B , i.e., $|\Phi_B|$, is increasing/decreasing:
 - 1.1 $|\Phi_B|$ is increasing if both $\frac{d\Phi_B}{dt}$ and Φ_B have the same sign (both positive or both negative)
 - 1.2 $|\Phi_B|$ is decreasing if the sign of $\frac{d\Phi_B}{dt}$ is different from the sign of Φ_B
2. If $|\Phi_B|$ is increasing/decreasing, the direction of I is such that the new $\mathbf{B}'$ it creates, wants to stop $|\Phi_B|$ from increasing/decreasing
3. The direction of $\mathcal{E}$ is the same as I
4. The sign of $\mathcal{E}$ is positive if $\mathbf{B}'$ is the direction of $\mathbf{A}$, negative if $\mathbf{B}'$ is the opposite direction of $\mathbf{A}$



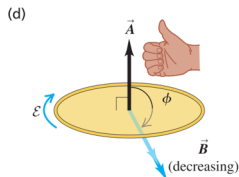
- Flux is positive ($\Phi_B > 0$) ...
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- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

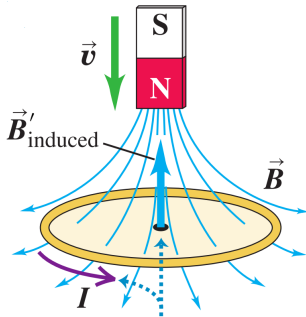
Based on change in Φ_B :

Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):

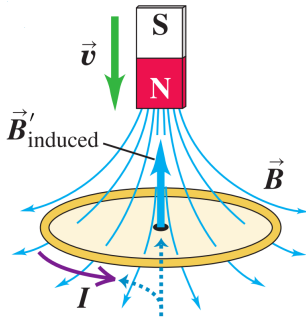


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$

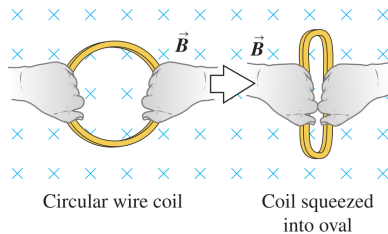


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):

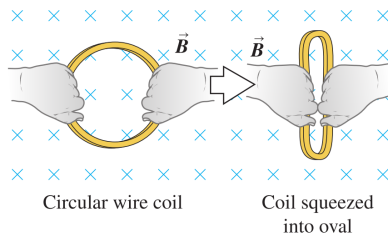


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$

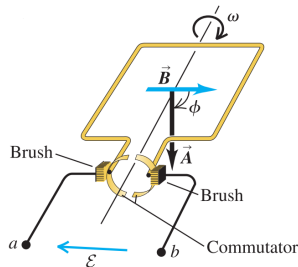


Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$
- If change in $|\Phi_B|$ is due to a change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other



Lenz's Law

How does $\mathbf{B}'$ created by induced $\mathcal{E}$ and its induced current I stop $|\Phi_B|$ from further increasing/decreasing?

Based on change in Φ_B :

- If change in $|\Phi_B|$ is due to a change in $\mathbf{B}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{B}$
- If change in $|\Phi_B|$ is due to a change in $d\mathbf{A}$ or $\mathbf{A}$ (magnitude, direction, or both):
 - I will be such that induced $\mathbf{B}'$ stops further change in $\mathbf{A}$
- If change in $|\Phi_B|$ is due to a change in $\mathbf{B} \cdot d\mathbf{A}$, i.e., the orientation of $\mathbf{B}$ and $d\mathbf{A}$ with respect to each other
 - I will be such that induced $\mathbf{B}'$ stops further change in the angle between $\mathbf{B}$ and $\mathbf{A}$

